

DETECTING WORKERS WITHOUT HELMET

Using YOLO V5 And MaixCam

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In industrial settings, safety helmets are a critical element of personal protective equipment (PPE), acting as the final barrier when other measures fail. Safety departments enforce PPE rules across process plants, especially among large numbers of contract workers. While training builds hazard awareness, studies show supervision plays a stronger role in accident prevention.

To address this need more effectively, a technology-led solution replaces manual oversight with an intelligent, machine-based system.

A high-fidelity camera continuously monitors the approach area. If a worker without a helmet is detected, the system triggers an instant alert—either a cautionary alarm or a verbal instruction to comply. A single camera can cover large zones, ensuring round-the-clock, impartial enforcement.

Once a violation is flagged, further action—such as activating an audio alert or controlling a gate—is handled by a microprocessor-based setup.

The solution uses MaixCam's GPIO library to manage these automated responses, delivering scalability and cost-efficiency. Acting as a tireless, unbiased supervisor, this AI-powered safety system reduces reliance on human monitoring while boosting compliance. A prototype is shown in Fig. 1.

The required components are listed in the Bill of Materials table. An LED or buzzer may be added for alerts, though not included in the Bill of Materials.

EFY note. Since the same board is used in this solution, refer to 'Face Recognition Using YOLO,' published in the April issue. That article provides additional details regarding the MaixCam development board,

including its usage and installation, which should be reviewed before proceeding.

Hardware setup

AI processing requires substantial computing power, and running complex models on a microcontroller board may not always be ideal. A Raspberry Pi 5 with 8GB RAM is a suitable choice, and it becomes even more powerful when paired with an AI processor HAT (13 or more TOPS). This is particularly important because models such as YOLO require significant computational resources.

This solution, however, uses MaixCam, which combines an NPU

BILL OF MATERIALS

Components	Quantity
MaixCam	1
5V DC power adaptor	1



Fig. 1: Author's prototype



Fig. 2: Various face repositioning